Automated Promo Scout for AI Microdramas

High-Velocity Trailer Extraction, Social Scroller Optimization & Quality Auditing

50 Episodes	10.36 Seconds	20s - 60s Bounds	$0.00 / Show
Parsed & Audited	Full Batch Run Speed	Strict TikTok Window	Free-Tier / Local AST

EXECUTIVE SUMMARY & PROBLEM CHOICE

Why Problem 1? Short vertical microdrama is Kuku's primary growth frontier. Today, human editors manually watch every episode to extract promos--a severe bottleneck as show output scales. This submission delivers 'Promo Scout AI': a modular, production-ready system that automates 20-60s trailer discovery across 50 production screenplays. It combines deterministic AST signal processing (guaranteeing exact timestamps and zero mid-sentence cuts) with an LLM layer that judges psychological hooks, and ranks promos for optimal social media ad spend.

1. Architecture: What We Gave to AI vs. Deterministic Code

A common failure in AI workflows is prompting LLMs with entire 50-episode scripts, causing hallucinated timestamps, truncated dialogue, and heavy API costs. We strictly split deterministic signal processing from semantic reasoning:

[Deterministic Code] Python AST & Rule Engine

- Screenplay AST Parser: Millisecond-accurate extraction.
- Constraint Enforcer: Strict 20s <= Duration <= 60s bounds.
- Boundary Snapping: Prevents cutting mid-sentence/dialogue.
- Quality Auditor: Flags pronoun ambiguity & low kinetics.
- Zero Hallucinations: Timestamps are mathematically verified.

[AI / LLM Layer] Groq / Gemini / OpenAI (Swappable)

- 3-Second Hook Evaluator: Detects high thumb-stop arousal.
- Cliffhanger Scorer: Rates unresolved suspense & cut peaks.
- Viral Synthesis: Writes punchy 1-line ad justifications.
- Model Agnostic: Runs Groq Qwen/Llama (300 t/s) or Gemini.
- Offline Graceful Fallback: Seamless local scoring backup.

2. Viral Scoring Formulation & Segment Selection

Every prospective promo window W = [t_start, t_end] satisfies 20.0s <= (t_end - t_start) <= 60.0s. The Composite Viral Index V(W) evaluates three orthogonal axes plus an audience acquisition funnel modifier:

V(W) = [0.40 * Hook(W) + 0.40 * Cliffhanger(W) + 0.20 * Coherence(W)] * 10 + Funnel_Weight(Ep)

* Hook Score (H, 40%): Evaluated on [t_start, t_start + 3.0s]. Measures physical shock (slaps, grabs, weapon draws), high-voltage delivery (venomous, furious, screaming), and dynamic close-up camera framing.
* Cliffhanger Score (C, 40%): Evaluated on [t_end - 4.0s, t_end]. Detects unresolved ultimatums, audio string swells, and cut-to-black.
* Standalone Coherence (S, 20%): Evaluates dialogic tension (protagonist vs antagonist) without needing previous episodes.
* Segment Discovery: Evaluates continuous stretches as well as 2-segment dramatic stitches (opening shock + final cliffhanger).

3. Release Strategy: Which Promos to Post First & Why

A promo's conversion rate on TikTok/Instagram Reels depends strictly on audience coldness. We implement a 3-Tier Funnel:

Tier 1: Cold Traffic (Ep 1-5)
POST ON LAUNCH DAY
Primal emotional hooks (altar abandonment, disowned by family, secret Duke). Requires zero lore; generates maximum outrage and drive-to-app installs.

Tier 2: Mid-Funnel (Ep 6-20)
POST IN WEEK 1-2
High-voltage romantic standoff and mystery cues (treating wounded Duke, poison threats). Ideal for retargeting viewers who saw Day 1 promos.

Tier 3: Fan Retention (Ep 21-50)
POST LATER IN CAMPAIGN
Intricate multi-character revelations and climax resolution. Withheld early to prevent spoiling the primary mystery for new users.

4. Production Run Results: Ranks #1, #2 & #3 Across All 50 Episodes

The automated pipeline evaluated all 50 episodes of 'I Married a Broken Soldier' in 10.36s. Below are the definitive Top 3 promos recommended for immediate media spend, detailed with timestamps, viral hooks, cut points, and conversion psychology:

#1	Episode 3 | Timestamps: 00:25.80 --> 01:06.00 (40.2s)	Score: 115.0 / 115.0 | Day 1 Top Budget Priority (Tier 1: Cold Traffic)
[3s Hook]	Guards violently yank Lady Juliana backward by the shoulders, tearing her from the pew while she screams 'Let me go!' in the middle of a sacred chapel.	
[Cliffhanger Cut]	Juliana is dragged toward the exit in a blur of motion, her plea swallowed by the stone walls, cutting to black right as an imposing shadow falls over the guards--freezing before the scarred warrior's intervention is shown.	
[Reason & Viral Why]	Primal physical aggression combined with the sacred sanctuary betrayal creates an immediate 'fight-or-flight' protective empathy spike on TikTok/Reels feeds. Universal outrage converts cold viewers with zero prior lore, driving immediate app installs.	

#2	Episode 4 | Timestamps: 01:54.00 --> 02:29.00 (35.0s)	Score: 108.0 / 115.0 | Day 1 Secondary Ad Creative (Tier 1: Cold Traffic)
[3s Hook]	Lady Ashcroft's venomous shout 'You are disowned!' paired with Gerald's furious recoil: 'For a penniless nobody!' instantly establishes acute betrayal and family conflict.	
[Cliffhanger Cut]	Juliana and the scarred warrior walk hand-in-hand out into the blinding sunlight, freezing on the entire noble family standing in speechless shock--leaving viewers desperate to know whether this 'beggar' is truly the hidden Duke.	
[Reason & Viral Why]	Hits the high-converting 'underdog revenge & secret billionaire/duke' trope. The contrast between cruel family disownment and Juliana's dignified, triumphant exit gives viewers immediate emotional gratification while opening an irresistible curiosity gap about the warrior's true identity.	

#3	Episode 1 | Timestamps: 01:23.70 --> 02:17.50 (53.8s)	Score: 102.0 / 115.0 | Launch Week Anchor Ad (Tier 1: Cold Traffic)
[3s Hook]	Lady Juliana's raw, trembling whisper 'Father... please look at me' is met with the priest's stone-cold refusal to make eye contact, instantly establishing visceral emotional cruelty.	
[Cliffhanger Cut]	Juliana's knuckles whiten as she clutches the silk folds of her dead mother's gown in suppressed fury while Gerald sneers that she is of 'low value' compared to Clarissa, cutting to black on an ominous, swelling string drone.	
[Reason & Viral Why]	Public altar abandonment is the core inciting incident of vertical drama. The emotional injustice of being cast aside for her sister in her deceased mother's wedding gown creates intense moral indignation that compels viewers to watch the full series for retribution.	

5. Quality Checking: Where Does the Tool Pick Badly & Mitigations

The brief explicitly asks: 'Where does your tool pick badly?' Rather than masking flaws, we built an automated Quality Auditor (quality_checker.py) that inspects every candidate promo for real-world microdrama failure modes:

1. Pronoun Ambiguity ('Cold Start' Failure)
Flaw: Clip opens mid-scene with 'He lied to you' when 'He' was introduced earlier. Scrollers bounce from lack of context.
Mitigation: Auditor detects opening pronouns; penalizes score by 8pts unless preceded by an explicit character action.

2. Low-Kinetic 'Talking Heads'
Flaw: High drama words ('poison', 'secret') spoken while actors stand motionless. Bounces on TikTok within 2s.
Mitigation: Visual kinetic filter enforces at least 2 camera/blocking shifts per window, deducting 10pts if static.

3. Expository Cut Transitions
Flaw: LLM occasionally cuts on trailing actions like 'Juliana walks out the door,' bleeding off dramatic momentum.
Mitigation: Cut-point snapping forces endings onto close-up reaction shots or peak audio swells before physical exit.

4. Tool Error Encountered & Resolved
Flaw: In Episode 6, headers were formatted in ALL CAPS ('SCENE HEADING:'), causing early regex to miss scenes.
Mitigation: Replaced strict matchers with case-insensitive tokenization and added Pydantic validation for JSON.

6. Real-World Scaling, Economics & Human-in-the-Loop

Human-in-the-Loop Workflow
- Editors never start from scratch. The tool serves the top 3 ranked cuts per episode with 1-click preview and timestamp sync.
- The editor reviews, clicks 'Approve', or nudges boundaries by +/- 2 seconds.
- Reduces promo hunting from 4 hours to 8 minutes per show.

Production Economics at Scale
- Local / Groq Tier Cost: $0.00 for the entire 50-episode run.
- Commercial LLM (GPT-4o-mini / Gemini Flash): < $0.03 per entire 50-episode show (~100k tokens total).
- Scaling to 50 new shows weekly costs under $1.50 per week, unlocking infinite ad variant testing.

DELIVERABLES INCLUDED: Interactive Web Dashboard (FastAPI+Tailwind) | CLI Tool | Full 50-Ep Raw Outputs (JSON/CSV) | Video Script